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Reliability & Risk Engineering

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Techno Economic Assessment of CO_2 Transport Resilience in the CCTS Chain under Main Connection Failure Risk

Semester Project

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Zürich, January 4, 2026

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Abstract

The deployment of Carbon Capture Transport and Storage (CCTS) infrastructure relies on reliable and cost-effective CO_2 transport systems linking emitters to geological storage sites. Transport interruptions can result in production curtailment or direct atmospheric release, making transport resilience an increasingly important design consideration as carbon pricing rises.

This study develops a techno-economic framework to assess the feasibility of resilience strategies in CO_2 transport networks. Feasibility is defined as the ability of a resilience option to achieve lower total system costs than directly emitting CO_2 in the atmosphere under a given set of operating conditions. The framework evaluates temporary CO_2 storage, redundant transport configurations, and direct emissions as a last resort option, accounting for failure durations, infrastructure utilization, dynamic carbon taxation, and operational emissions.

The analysis investigates how transport distance, annual CO_2 mass flow of the Waste to Energy (WtE) power plant, and duration time for the failure period of the main connection influence resilience feasibility. The results show that resilience is generally not cost effective under low utilization or infrequent failures, but becomes economically preferable under prolonged disruptions and sufficiently high system utilization. Temporary storage emerges as the most robust and cost effective resilience option across a wide range of conditions, while container based transport provides a viable alternative in systems with intermediate and low transport distances.

The framework is applied to a case study connecting the Hagenholz WtE facility to the Northern Lights storage site. The results demonstrate that network configuration strongly affects resilience feasibility and highlight the conditions under which resilient transport strategies can deliver both economic and environmental benefits.

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List of Acronyms

CAPEX Capital Expenditure

OPEX Operational Expenditure

CCTS Carbon Capture Transport and Storage

CRF Capital Recovery Factor

USD United States Dollar

EUR Euro

ETH Eidgenössische Technische Hochschule (ETH Zürich)

RRE Reliability and Risk Engineering

UC Unitary Cost

LC Levelized Cost

WtE Waste to Energy

NPV Net Present Value

CAPEX Capital Expenditure

OPEX Operational Expenditure

TEA Techno Economic Analysis

List of Symbols

Latin letters

Symbol	Units	Description
i	-	Interest rate (fraction per year)
L	km	One-way transport distance
D_b	Days	Buffer duration
$M_{\mathrm{CO_2}}$	t	Annual CO_2 capacity
M_{b}	t	Capacity of CO2 during buffer period
$M_{\mathrm{CO_{2f}ail}}$	t	Capacity of CO2 during failure for a year
P_{diesel}	€/L	Diesel price
c_f	L/100 km	Average fuel consumption
v	km/h	Average speed
c_m	€/km	Maintenance cost coefficient
c_t	€/km	Tire cost coefficient
c_i	€/km	Insurance cost coefficient
$C_{\mathrm{driver,h}}$	€/h	Driver hourly wage
Cap_{truck}	t	Truck CO_2 capacity
t_{load}	h	Loading time
t_{unload}	h	Unloading time (train)
t_{border}	h	Border delay
t_{break}	h	Driver rest time
$C_{\mathrm{truck,veh}}$	USD	Truck cost
C_{trailer}	USD	Trailer cost
n	years	Truck lifetime
N_{wagon}	–	Wagons per train
Cap_{wagon}	t	Wagon CO_2 capacity
C_{loco}	USD	Locomotive cost
C_{wagon}	USD	Wagon cost
C_{OPEX}	€/train·km	OPEX per train-km
m_{maint}	–	Annual maintenance fraction
n_L	years	Locomotive lifetime
n_W	years	Wagon lifetime

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1 Introduction

Achieving deep reductions in carbon dioxide (CO_2) emissions is central to international climate mitigation efforts. In line with the objectives of the Paris Agreement, the Swiss Federal Council has committed to limiting global temperature increase to 1.5 °C by pursuing net-zero greenhouse gas emissions by 2050 (*Goals and strategies of Swiss climate policy* 2025). Reaching this target will require a portfolio of mitigation measures across multiple sectors, among which carbon capture, transport, and storage (CCTS) is expected to play an important role. In May 2022, the Swiss Federal Council approved measures to support the deployment of CCTS technologies by mid-century, highlighting the strategic relevance of transport and storage infrastructure for national de-carbonization pathways (*Roadmap to capture, remove and store carbon* 2025).

While large-scale CO_2 capture and geological storage technologies have advanced substantially, the transport segment of the CCTS chain is increasingly recognized as a critical point of vulnerability. CO_2 transport infrastructure and particularly pipelines, is typically designed for continuous steady state operation, with limited flexibility to accommodate unplanned interruptions. Even short outages can require curtailment of capture or result in direct atmospheric release of captured CO_2 if no alternative handling options are available (Budinis et al. 2018). As transport distances increase and volumes scale up, this reliance on uninterrupted operation introduces a systemic risk that affects the performance of the entire CCTS chain.

Transport failures are particularly consequential from a techno-economic perspective. Conventional techno-economic analyses of CCTS systems generally assume full availability of transport infrastructure over the system lifetime, implicitly spreading capital and operating costs over uninterrupted annual operation (Oeuvray et al. 2024; Gabrielli et al. 2022). Under this assumption, the economic and environmental consequences of transport outages are not internalized in levelized cost estimates. In practice, however, infrastructure failures can be costly, with repair times ranging from days to weeks, as observed in oil and gas transmission networks (Zakikhani et al. 2020). In a CO_2 transport network, such outages transform a temporary technical failure into an immediate trade-off between cost and emissions. Operators must either absorb the cost of alternative transport options or accept temporary venting of captured CO_2. As carbon pricing increases, the economic penalty associated with venting grows, directly linking transport reliability to long term system costs and climate outcomes.

This interplay between infrastructure reliability, economic performance, and emissions raises a fundamental question that remains insufficiently explored in existing CCTS planning frameworks:

> *Under which conditions does it become economically rational to invest in transport resilience rather than accept temporary CO_2 emissions during failures?*

In this work, resilience is defined as the ability of a CCTS chain to maintain operational continuity during transport disruptions, either through redundant transport capacity (e.g. trucks, trains, ships, barges, or pipelines) or through temporary CO_2 storage that buffers captured mass until the primary transport corridor is restored. Comparable resilience concepts are well established in electricity systems and natural gas supply networks, where redundancy becomes cost effective only once the durations of the interruption, the scale of the system, and the utilization exceed certain thresholds (Carvalho et al. 2014; Cicilio et al. 2021). For CO_2 transport systems, however, such thresholds have not yet been systematically identified, and transport outages are rarely integrated into long term techno economic planning.

This study addresses this gap by developing a techno economic modeling framework that evaluates transport resilience as a conditional investment decision rather than a default design feature. The framework explicitly couples annual CO_2 mass flow of WtE power plants, transport distance, failure duration, and the temporal evolution of carbon pricing to identify break even conditions under which

resilience options become economically preferable compare to directly emitting CO_2 in the atmosphere. By shifting the focus from how resilience should be implemented to when it becomes rational to do so, the analysis highlights the dynamic nature of resilience feasibility over both system scale and time.

Through this formulation, the study establishes a basis for resilience aware CO_2 transport planning. The results demonstrate that redundancy is neither universally beneficial nor universally avoidable. Instead, its value emerges only under specific operating conditions that can be quantified. These insights provide guidance for designing CCTS systems that remain both economically efficient and environmentally robust under realistic infrastructure constraints.

2 Methodology

2.1 Modelling of resilience options

This section presents the Techno Economic Analysis (TEA) framework used to evaluate the performance and cost of different CO_2 transport resilience strategies. The model combines engineering cost relationships with time dependent operational assumptions to quantify how disruptions in the permanent connection affect system requirements and total cost per ton of CO_2. The analysis distinguishes between normal operation, supported by a pipeline based permanent connection, and a failure window during which the connection becomes unavailable and alternative solutions must be deployed.

Within this framework, two resilience strategies are assessed, which are a Backup Connection transport option and a Temporary Storage option, which installs sufficient capacity to buffer the disrupted flow until full operation resumes, the two options are operating only during failure periods of the main connection of the system. Both approaches are evaluated through annualized formulations Capital Expenditure (CAPEX) and Operational Expenditure (OPEX) and compared against a baseline case of direct emissions. The model takes as inputs the total length of the main connection, the total CO_2 mass flow rate of the WtE power plant that needs to be transported, the expected failure period of the main connection, while also the interest rate.

In the event of a failure of the main connection of the CCTS chain to the permanent storage site, several alternative transport options are considered to ensure the continuity of CO_2 the supply chain. The backup solutions examined in this study include container trucks, trains, barges, ships, and both onshore and offshore pipelines. Each option is evaluated through a dedicated cost model that reflects its technical characteristics and operational behavior during failure periods.

In (Oeuvray et al. 2024), the unit cost expressions for truck and train transport are formulated for regular operation and do not explicitly depend on transported mass. This assumption is appropriate for primary transport services operating at high utilization, where economies of scale can be fully realized. In the context of transport resilience, however, backup options are activated only during failure events and therefore operate for a limited number of hours per year. Under such conditions, technologies that rely on economies of scale are penalized rather than favored, as their fixed and capacity driven costs are distributed over small transported volumes, resulting in disproportionately high costs per ton.

In (Oeuvray et al. 2024), the unit cost expressions for truck and train transport are formulated for regular operation and do not explicitly depend on transported mass. Under the low operating hours of the resilient strategies, economies of scale cannot be fully exploited. Therefore, to ensure consistency across transport modes under backup operation, the truck and train models used in this work introduce an explicit dependence on transported mass, while remaining aligned with the structure and assumptions of (Oeuvray et al. 2024). Additional parameters to complete the model come from (Myers et al. 2024).

For pipelines, barges, and ships, the cost formulations are adopted directly from (Oeuvray et al. 2024). Although these modes are likewise considered as backup options in this study, their cost models already incorporate mass dependent scaling and capacity utilization effects. This ensures that all transport options reflect the cost impact of low utilization under backup operation, enabling a consistent comparison of resilience strategies.

The two metrics that the models are able to provide are the annualized CAPEX and OPEX per ton of CO_2 considering the total period during which the main connection is not operational.

For the annualization of the CAPEX of each connection, the Capital Recovery Factor (CRF) is used, which is presented in Eq. 1, where n is the expected lifetime of each connection j and i is the applied interest rate, which is set at 7%, which is a common value used in literature for CCTS projects (Oeuvray et al. 2024).

$$CRF = \frac{i(1+i)^{n_j}}{(1+i)^{n_j} - 1} \tag{1}$$

2.1.1 Dedicated truck transport model

A detailed description of the development of a truck-based model designed to provide transport capacity during failures of the main connection in the chain is presented in this section. The model accounts for the CO_2 volumes requiring transport in each failure period, including low volume scenarios associated with short disruptions, by determining the required number of trucks and the corresponding CAPEX and OPEX.

For the structure of the model the time for the total trip should be calculated using Eq. 2, since each truck performs round trips between the total length of the connection. The operational cost parameters of each trip are defined from Eq. 42 presented in Appendix A, which includes the cost of tires, fuel, maintenance, insurance, and driver wages. The hourly cost of a truck driver is derived from an annual salary of $55,000, adjusted by a factor of 0.86 to transform US dollars to Euros. This annual cost is then distributed over a typical working schedule of 260 days per year and eight working hours per day. Dividing the adjusted annual cost by the total number of working hours yields the corresponding hourly driver cost.

$$t_{\text{trip}} = 2t_{\text{load}} + t_{\text{border}} + t_{\text{break}} + \frac{2L}{v} \text{(Oeuvray et al. 2024)[h]} \tag{2}$$

Each failure is planned ahead and assumed to last a specific amount of days, which are the buffer days (D_b). The operational time available during the buffer period is defined by Eq. 3. In addition, the total number of round trips for each truck is calculated using Eq. 4, where the time for loading, unloading and the total time to cover the required distance are taken into account.

$$t_{\text{avail}} = D_b \times 8 \quad \text{[h]} \tag{3}$$

$$N_{\text{trips}} = \max\left(1, \left\lfloor \frac{t_{\text{avail}}}{t_{\text{trip}}} \right\rfloor\right) \text{(Oeuvray et al. 2024)} \tag{4}$$

For the analysis and the calculation of the total cost per mass of CO_2, the mass of CO_2 that it is to be transported during the failure period needs to be calculated and therefore Eq. 5 is to be used. The buffer days are used for this calculation and therefore the total annual mass of CO_2 from the power plant (M_{CO_2}) can be converted into total capacity for a specific time period. Assuming a single failure per year, the total annual CO_2 transport requirement during failure conditions is equal to M_b.

Thus, total number of trucks required for the transport is given by Eq. 6, which states that the number of trucks must be the smallest integer that satisfies the two conditions presented.

$$M_b = M_{CO_2} \frac{D_b}{365} \text{[t]} \tag{5}$$

$$N_{\text{truck}} = \min\left\{ n \in \mathbb{Z} \,\middle|\, n \geq 1 \text{ and } n \geq \frac{M_b}{N_{\text{trips}} Cap_{\text{truck}}} \right\} \tag{6}$$

To calculate the annualized CAPEX and OPEX per ton of CO_2 transported during the failure period, the total investment cost is first determined using Eq. 7. The annual maintenance cost is then calculated by applying the operational fraction for trucks in Eq. 8. Finally, the total operational cost is divided by the transported mass during the failure period to obtain the OPEX per ton (Eq. 9).

$$I_C = N_{\text{truck}}(C_{\text{truck,veh}} + C_{\text{trailer}})[€] \tag{7}$$

$$OC_{\text{failure}} = N_{\text{truck}} N_{\text{trips}} \Big[C_{\text{fuel}} + C_{\text{maint}} + C_{\text{tires}} + C_{\text{ins}} space[€] + t_{\text{trip}}\, C_{\text{driver,h}} \Big][€] \tag{8}$$

$$OPEX_{\text{per ton}} = \frac{OC_{\text{failure}}}{M_b}[€/\text{t}] \tag{9}$$

In terms of capital expenditure per ton ($CAPEX_{\text{per ton}}$), it must first be annualized using the capital recovery factor presented in Eq. 1 and then divided by the total mass that goes through during the failure period (Eq. 10).

$$CAPEX_{\text{per ton}} = \frac{CRF\, I_C}{M_b}[€/\text{t}] \tag{10}$$

2.1.2 Dedicated train transport model

As described in section 2.1, a dedicated train model is to be developed, with this section describing the detailed development of a mathematical model to calculate the annualized CAPEX and OPEX of trains operating exclusively during failures of the main connection, designed to transport the total CO_2 mass produced by the power plant during the corresponding failure period.

Each train configuration is assumed to consist of 20 container wagons, with each wagon providing a transport capacity of 50 t of CO_2. The total amount of CO_2 that a single train can carry in one trip is therefore obtained by multiplying the number of wagons by their individual capacity, resulting in a maximum train transport capacity of 1000 t per train. The travel time for each train can be calculated using Eq. 11, which takes into account the time to load and unload CO_2 as well as the time required for the round trip for the train.

$$t_{\text{trip,train}} = t_{\text{load}} + t_{\text{unload}} + \frac{2L}{v}[\text{h}](\text{Oeuvray et al. 2024}) \tag{11}$$

As for the fleet sizing, the available operation time during disruption is with the assumption that trains are able to operate throughout the entire day, and therefore the total number of operating hours available during the failure period is determined by multiplying the buffer duration by 24, since each buffer day provides a full day of operational time. Upon analysis, the total number of trips per train is then calculated using Eq. 12, which takes into account the available time and the time needed for each train to complete the trip, with the minimum number of trips being at least one. Using the results of Eq. 12 the number of trains needed is calculated using Eq. 13, where as for the trucks the number of trains is the smallest integer that satisfies the conditions presented.

$$N_{\text{trips,train}} = \max\left(1, \left\lfloor \frac{t_{\text{avail}}}{t_{\text{trip,train}}} \right\rfloor\right) \text{(Oeuvray et al. 2024)} \tag{12}$$

$$N_{\text{train}} = \min\left\{ n \in \mathbb{Z} \,\middle|\, n \geq 1 \text{ and } n \geq \frac{M_b}{N_{\text{trips,train}} Cap_{\text{train}}} \right\} \tag{13}$$

With the total number of trains calculated, the total operational cost per ton for each failure period is then calculated by Eq. 18, which divides the total operational cost calculated by Eq. 17, with the total mass that must be transported in the period equal to the buffer days. Regarding the calculation of the operating cost, the investment cost must be calculated and then the annual maintenance cost should be applied, which is a fraction of the total Investment cost calculated in Eq. 14.

$$I_C = N_{\text{train}}(C_{\text{loco}} + N_{\text{wagon}} C_{\text{wagon}})[€] \tag{14}$$

$$OC_{\text{var}} = N_{\text{train}} N_{\text{trips,train}} (2L) C_{\text{OPEX}} \tag{15}$$

$$OC_{\text{maint}} = m_{\text{maint}}\, I_C \tag{16}$$

$$OC_{\text{failure}} = OC_{\text{var}} + OC_{\text{maint}}[€] \tag{17}$$

$$OPEX_{perton} = \frac{OC_{\text{failure}}}{M_b}[€/\text{t}] \tag{18}$$

For the annualized CAPEX per ton, CRF presented in Eq. 1 is then used to convert the total investment cost to annual investment and then divide the total annualized IC cost by the total mass of CO_2 that is transported during the failure period using Eq. 19.

$$CAPEX_{perton} = \frac{N_{\text{train}}\left(CRF_L\, C_{\text{loco}} + N_{\text{wagon}} CRF_W\, C_{\text{wagon}}\right)}{M_b}[€/\text{t}]. \tag{19}$$

2.1.3 Barge, ship, and pipeline transport

This section presents the cost models implemented for CO_2 transport via barges, ships, and pipelines. The models presented in (Oeuvray et al. 2024), are well suited for the implementation that this study requires, considering the fact that the models already take into account for transported mass and are therefore suitable for representing non-continuous operation.

Each type of connection is characterized by a set of parameters a_1–a_4 that define the unit cost correlation, as shown in Tables 5–7 in Appendix B. The annualized CAPEX and OPEX per ton of CO_2 are calculated using the coefficients presented and the OPEX fraction. Since the mass dependent parameters in the model are tons per year, a flow should be used, and therefore the total annual mass of the WtE power plant is used M_{CO_2}.

The total levelized cost per ton of CO_2 ($LC_{\text{transport}}$) is calculated using Eq. 22, where the unit cost is multiplied by the total length of the connection, where the unit cost depending on the type of connection is given by Eq. 20 and Eq. 21.

- For **barges and ships**:

$$UC = a_1 + \frac{a_2}{L} + \frac{a_3}{M_{\mathrm{CO_2}}} [€/\mathrm{t{\cdot}km}] \tag{20}$$

- For **pipelines**:

$$UC = a_1 + a_2 \left(\frac{L}{d_{\mathrm{ref}}}\right)^{a_3} \left(\frac{M_{\mathrm{CO_2}}}{m_{\mathrm{ref}}}\right)^{a_4} [€/\mathrm{t{\cdot}km}] \tag{21}$$

$$LC_{\mathrm{transport}} = UC \times L[€/\mathrm{t}] \tag{22}$$

The total cost per ton is decomposed into capital and operational components using the OPEX fraction. This fraction represents the share of the annualized technology cost that is attributed to operational expenditures. Consequently, the operational cost is obtained by multiplying the levelized cost of the technology by the OPEX fraction, while the remaining share corresponds to the capital component. The $LC_{\mathrm{transport}}$ result assumes a much higher CO_2 throughput than actually achievable, because it divides the total CAPEX by the annual mass flow. In reality, for a resilient option, only the buffer capacity determines the amount of CO_2 that can pass during an outage. Therefore, the cost must be adjusted accordingly, as shown in Eq. 24, which represents the cost per ton when only one failure occurs per year with a duration equal to the buffer capacity.

$$OPEX_{\mathrm{per\ ton}} = LC_{\mathrm{transport}} \times f_{\mathrm{OPEX}}[€/\mathrm{t}] \tag{23}$$

$$CAPEX_{\mathrm{per\ ton}} = (LC_{\mathrm{transport}} - OPEX) \times \frac{M_{\mathrm{CO_2}}}{M_{\mathrm{b}}}[€/\mathrm{t}] \tag{24}$$

2.1.4 Model of temporary storage

According to the purpose of this study, for the resilience in CCTS network, temporary storage is considered to buffer the mass of CO_2, during the failure period. A detailed explanation of the methodology used to size the temporary CO_2 storage facility based on the required buffer duration and the estimation of its capital and operational costs using a capacity based scaling approach is presented.

The total storage capacity (in tons) corresponding to the duration of the buffer is determined by Eq. 5 as in all previous models.

The total investment cost of the temporary storage facility is estimated using a power-law scaling relation, expressed as

$$C_{\mathrm{IC}} = C_{\mathrm{ref}} \left(\frac{M_{\mathrm{b}}}{S_{\mathrm{ref}}}\right)^{n} \tag{25}$$

In this formulation, the reference cost is taken as $C_{\mathrm{ref}} = 14{,}350{,}000$ € (*Components of CCS Infrastructure - Interim CO Holding Options* 2025), corresponding to a reference storage capacity of $S_{\mathrm{ref}} = 5{,}100$ t. The scaling exponent is set to $n = 0.7$, following the recommendation of (Moch et al. 2022), who note that economies of scale are inherent to CCTS infrastructure. Specifically, they report that capital costs for CCTS facilities typically scale with the n-th power of capacity, with empirical estimates placing n in the range of 0.6–0.8 depending on the component under consideration. As buffer storage constitutes one of the cost relevant elements within an integrated CCTS chain, a conservative mid range value of $n = 0.7$ is adopted in this study for scaling storage-related capital expenditures.

The investment is CAPEX annualized over the storage life l and is categorized using Eq. 1 for a useful life of 20 years and a failure capacity equal to buffer days capacity.

$$CAPEX_{\text{per ton temp}} = \frac{CRF \times C_{\text{IC}}}{M_b} [€/\text{t}] \tag{26}$$

The annual operational cost is calculated as a fixed fraction of the total investment cost, where $f_{\text{OPEX}} = 0.02$ (i.e., 2% of the initial capital cost). (Life Cycle Engineering and Process & Equipment Development Corporation 2024)

$$OPEX_{\text{per ton temp}} = f_{\text{OPEX}} \times C_{\text{IC}} [€/\text{t}] \tag{27}$$

2.1.5 Temporary storage with oversized pipeline connection

The temporary storage configuration represents a resilience strategy in which the CO_2 that cannot be transported during a pipeline outage is accumulated in a dedicated storage facility and subsequently transferred to the permanent storage site using the main pipeline once normal operation resumes. This option is evaluated only for systems whose permanent connection is a pipeline. The underlying assumption is that the marginal investment required to increase pipeline capacity is relatively small, making it economically feasible to oversize the pipeline so that buffered CO_2 can be gradually discharged after the failure period. Additionally, for transport with trucks, trains, barges or ships, temporary storage is not applied, because they will already require the use of an additional transport unit for the discharge of the buffered mass. Consequently, the CO_2 can be transferred directly, eliminating the need for temporary storage and its associated costs.

The total cost of the strategy consists of two main components. First, the storage facility itself must be constructed with sufficient capacity to handle the interrupted flow. Its CAPEX and OPEX are calculated from the model that was described in section 2.1.4

The permanent pipeline must be large enough for the stored CO_2 to be evacuated once the system returns to normal operation. Because the buffer is not released instantaneously but rather progressively discharged and since no more than one failure is expected per year, the model assumes that the buffer capacity is distributed throughout an entire year. To quantify the financial implications of oversizing, the cost model is applied twice, once to determine the annual cost of the baseline pipeline and once to determine the annual cost of the oversized variant. The difference between these two calculations represents the annual incremental $\text{CAPEX}_{\text{per ton}}$ and $\text{OPEX}_{\text{per ton}}$ attributable to the temporary storage strategy. These incremental amounts are expressed in euros per ton and account solely for the additional capacity required to clear the buffered inventory.

The additional annualized capital cost per ton is calculated by Eq. 28, the model used for those calculations is the one presented in section 2.1.3, for pipelines with the total annual flow for the oversized (over) case being $M_{\text{CO}_2} + M_{\text{b}}$ per year and M_{CO_2} per year in the normal operating pipeline. The total CAPEX per ton of both cases is then multiplied by their corresponding mass that yields annual capital costs, and dividing their difference by M_{b} expresses the oversizing effect as an additional capital cost per ton of disrupted CO_2.

$$CAPEX_{\text{rel, per ton}} = \frac{CAPEX_{\text{over, per ton}} \left(M_{\text{CO}_2} + M_{\text{b}}\right) - CAPEX_{\text{main, per ton}} \left(M_{\text{CO}_2}\right)}{M_{\text{b}}} [€/\text{t}] \tag{28}$$

$$OPEX_{\text{rel, per ton}} = OPEX_{\text{over, per ton}} - OPEX_{\text{main, per ton}} [€/\text{t}], \tag{29}$$

In Eq. (28), the terms $CAPEX_{\text{over}}$ and $OPEX_{\text{main}}$ are specific capital costs (per ton) for the oversized and baseline pipelines, respectively. The quantity $CAPEX_{\text{rel}}$ and $OPEX_{\text{rel}}$ is the incremental specific capital and operational cost per ton associated with moving from the baseline to the oversized pipeline configuration.

The total cost per ton of buffered CO_2 for the temporary storage strategy for a failure period of failure days combines the contribution of the storage facility with the additional costs induced by pipeline oversizing. The capital and operational components are given by

$$CAPEX_{\text{per ton}} = CAPEX_{\text{per ton temp}} + CAPEX_{\text{rel, per ton}}[€/\text{t}], \tag{30}$$

$$OPEX_{\text{per ton}} = OPEX_{\text{per ton temp}} + OPEX_{\text{rel, per ton}}[€/\text{t}], \tag{31}$$

In Equations 30 and 31 $\text{CAPEX}_{\text{per ton temp}}$ and $\text{OPEX}_{\text{per ton temp}}$, represent the cost calculated for temporary storage in section 2.1.4. The relative terms $CAPEX_{\text{rel, per ton}}$ and $OPEX_{\text{rel, per ton}}$ then add the incremental per ton cost due to pipeline oversizing, yielding a consistent measure of capital and operational cost per ton for the Temporary Storage resilience option.

2.2 Direct emissions

The costs of all previous models are going to be compared to a benchmark case, where no resilience is chosen and therefore the entire CO_2 mass is directly emitted into the atmosphere.

In the case of direct emissions, the associated cost corresponds to the carbon tax that the WtE power plant must pay, minus the relative costs that are not applicable under these conditions. For the calculation of the carbon price on CO_2 in the next years, the study of (Mantulet et al. 2023) was used, at which the price is expected to be stable until the year 2030 due to policies that are currently affecting its price and afterward an exponential increase is expected. Figure 1 shows the evolution of the tax using as the starting price the one of 2025 in Europe, which is 78 [€/t] (*EU Carbon Permits - Price - Chart - Historical Data - News* 2025). This graph replicates the result of the research of (Mantulet et al. 2023) which is based on the Enerdata 2023 model (*Research on energy efficiency, CO2 emissions, energy consumption, forecast* 2025)

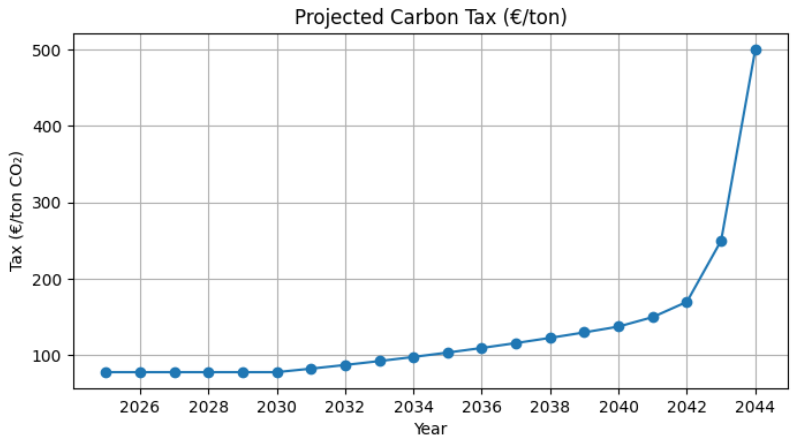


Figure 1: Tax evolution until year 2044

Equation 32 represents the operational cost for directly releasing CO2 in the atmosphere, while there is no capital expenditure associated with this option and therefore $CAPEX_{perton} = 0$.

$$OPEX_{\textbf{per ton}} = T - OPEX_{\textbf{carbon capture}} - C_{\textbf{permanent storage}}[€/\text{t}] \tag{32}$$

where the term T denotes the carbon tax expressed in €/t. The operational cost of the carbon capture unit is calculated following the approach in (Røsjorde and Carpenter n.d.), where it is assumed that the

OPEX is a fixed fraction of the capital cost of carbon capture. Specifically,

$$OPEX_{\text{carbon capture}} = CAPEX_{\text{carbon capture}} \times 0.05[€/\text{t}],$$

with a capital cost of $CAPEX_{\text{carbon capture}} = 62$ €/t as reported in (*Why the Cost of Carbon Capture and Storage Remains Persistently High — International Institute for Sustainable Development* 2025). In addition, the cost of permanent geological storage is taken as 13 €/t, consistent with the value provided in (*The cost of subsurface storage of CO2* 2025). Together, these elements define the baseline costs that must be taken into account when evaluating the economic implications of direct emissions during failure periods.

2.3 Discounting of future costs and accounting for main connection

This section outlines the cost methodology for resilience options, accounting for non-operational permanent connections during failures and discounting future costs to present value to reflect the time value of money.

For resilience options, the operational cost associated with the permanent connection ($\text{OPEX}_{\text{per ton perm}}$) must be subtracted from the resulting cost per ton. During a failure event, the permanent connection is not operating and therefore when the resilience options are considered there is no OPEX associated with the permanent connection, thus ensuring that only the additional cost attributable to the backup system is accounted for. The operational cost of the permanent connection is calculated in the same way as presented in the section backup connections (Sec. 2.1).

$$OPEX_{\text{resilince,j}} = OPEX_{\text{per ton}_j} - OPEX_{\text{per ton perm}}[€/\text{t}] \tag{33}$$

For all capital and operational costs calculated in the following subsections, the CAPEX must be scaled compared to the total failure days expected in a year (Eq. 34). The connection is being built to allow for capacity equal to the buffer days, but it might fail once every multiple years, instead of every year. Therefore, this must be a factor that must be taken into account in the calculations. The final CAPEX will be taken from Eq. 35.

$$M_{\text{CO}_2\text{fail}} = M_{\text{CO}_2} \times \frac{FailureDays}{365}[\text{t}] \tag{34}$$

$$CAPEX_{\text{resilince,j}} = CAPEX_{\text{per ton,j}} \frac{M_{\text{b}}}{M_{\text{CO}_2\text{fail}}}[€/\text{t}] \tag{35}$$

To ensure economic consistency across all scenarios, all future cost values were converted to their present value (PV) in the base year using a constant discount rate of $r = 7\%$. This approach enables a direct comparison of alternative investment timings and technology options on a common monetary basis.

For each simulated year t, the total annual cost per ton of CO_2 that includes capital recovery, operation and maintenance, and any applicable carbon tax, was discounted to present value according to:

$$C_{t,\text{PV},i} = \frac{CAPEX_{resilience,j} + OPEX_{resilience,j}}{(1+r)^{(t-t_0)}}[€/\text{t}] \tag{36}$$

where t_0 is the reference year (2025) and j is the option under consideration. This formulation ensures that all future monetary flows are expressed in terms of their current economic worth, consistent with standard techno–economic evaluation practices.

Importantly, the carbon tax was treated as a nominal annual cash flow and was not discounted separately before entering the cost model. This avoids double discounting effects, since the tax is already embedded in the total annual cost. Hence, only the resulting total cost per ton (C_t) was discounted once, at the final stage of the calculation.

In practical terms, this means that for each scenario, the full cost of transport and storage in year t was first evaluated in nominal euros using the techno economic model and the corresponding tax level for that year, and then discounted to present value using the expression above. The resulting value $C_{t,\mathrm{PV}}$ represents the total cost per ton of CO_2 in today's euros, allowing a consistent comparison of investment decisions across different years.

2.4 Emissions calculation

This section calculates the total emissions of the different resilience options, as well as the Direct Emissions using a simplified model, based on Life Cycle Assessments (LCA) of the different technologies.

Using the failure mass, emissions for every transport mode including trucks, trains, barges, ships, and pipelines are evaluated using the same generic formulation:

$$E = Emissionfactor_j \, L \, M_{\mathrm{CO2,fail}}, \tag{37}$$

where $Emissionfactor_j$ denotes the specific emission factor for transport mode j, for the emissions factors the results of the paper by (Nöhl et al. 2025) are used by replicating the emissions factors of each technology throughout the years, the total variation of the emission factors is presented in figure

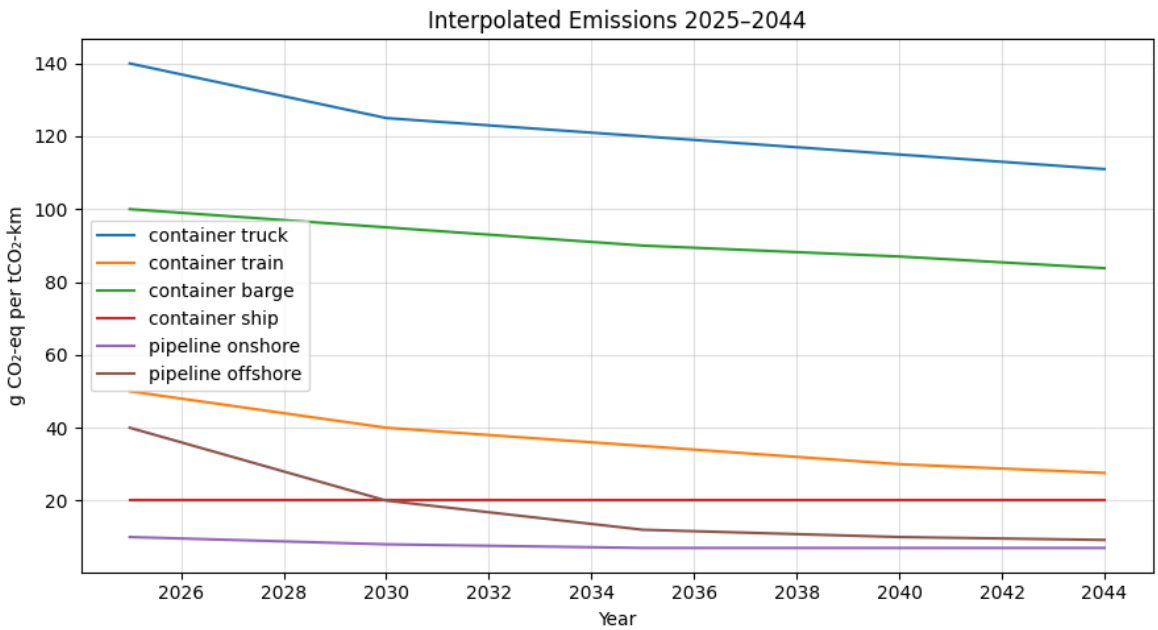


Figure 2: Emission factors from 2025 until 2044

The Temporary Storage resilience option also follows the same emission model, since the CO_2 accumulated during a failure event is ultimately transported through the same oversized permanent pipeline designed to handle both the nominal flow and the delayed release of stored CO_2. Thus, Temporary Storage does not alter the transport mechanism or the resulting emissions, it only changes the timing of when the failure mass enters the system. This assumption ensures low emissions for the temporary

storage system and provides a useful benchmark for comparison with the direct emissions case. However, it is not realistic.

Finally, for the Direct Emissions case, the total emissions correspond to the full mass of CO_2 generated during the failure period, as no transport occurs and the entire failure mass is released directly to the atmosphere.

2.5 Hagenholz case study

This subsection applies the proposed resilience modeling framework to the Hagenholz WtE facility as a representative case study. The objective is to translate the generic techno economic formulation into a system specific assessment that reflects realistic transport distances, operating conditions, and infrastructure reliability

In the original model, the Failure time [Days] was treated as an independent input and explored through discrete outage durations. In the revised reliability based framework, outage durations arise endogenously from the technology specific failure rate and the buffer capacity required for uninterrupted operation. This approach follows (Gabrielli et al. 2022), where infrastructure interruption frequencies replace exogenously assigned downtime windows.

Each transport technology is assigned a failure rate λ (failures per ton per kilometer), taken from the ranges reported in (Gabrielli et al. 2022). For a transport distance L and annual mass produced by the power plant ($M_{\mathrm{CO_2}}$), the expected CO_2 mass not transported, due to failures is computed as:

$$M_{\mathrm{CO2,fail}} = \lambda\, L\, M_{\mathrm{CO_2}}. \quad (38)$$

For this case study, the amount of CO2 per year for the Hagenholz WtE facility is 0.264 [$MtCO_2$/y] (Federal Office for the Environment (FOEN) 2025).

Gabrielli et al. (Gabrielli et al. 2022) also report typical storage autonomy requirements (*buffer days*), representing the minimum duration that must be covered locally during an interruption. These are adopted directly and listed in Table 2.

Table 2: Failure rates and buffer-day requirements adopted from Gabrielli et al. (Gabrielli et al. 2022).

Transport mode	**Buffer days B [days]**	**Failure rate λ [1/(t km)]**
Container truck	2	1.0×10^{-4}
Container train	1	1.0×10^{-4}
Container barge	10	1.0×10^{-5}
Container ship	10	1.0×10^{-5}
Pipeline onshore	20	1.3×10^{-4}
Pipeline offshore	20	1.3×10^{-4}

This reliability formulation removes arbitrary outage durations and replaces them with a technology and distance dependent estimate consistent with realistic operational behavior.

In the general model formulation the main transport connection was initially assumed to be an onshore pipeline. This assumption ensured consistency with temporary storage, which is only feasible when the main line can be oversized to provide buffering capacity.

In the case study formulation the main connection is treated explicitly as an input, allowing the operational cost of the underlying transport mode, to be incorporated into the total cost of each resilience option. Temporary storage remains available exclusively when the main connection is a pipeline.

To guarantee physical and operational feasibility, each main connection type is associated with a restricted set of resilience or alternative technologies. The complete feasibility mapping can be found in Appendix C, where all the resilience options are listed for each main connection type.

The physical structure of the case study route follows (Burger et al. 2024). The corresponding map is included in Figure 3, which illustrates the sequence of connections that link the Hagenholz WtE plant to the Northern Lights storage site. This map defines the spatial structure on which all failure modeling, technology filtering, and resilience analysis are applied. The detailed route is presented in Appendix C, with a detailed analysis of both the connection and the distances.

Figure 3: Main-connection layout used in the case study, based on (Burger et al. 2024).

Based on the same distances, an alternative scenario was assessed in which all main connections are assumed to be pipelines. This choice is supported by several studies that highlight the need for a European CO_2 pipeline backbone to enable large scale CCS deployment. Research in (Stewart et al. 2014) shows that an integrated cross border CO_2 pipeline network is technically feasible and cost effective, noting that an extensive pipeline infrastructure is required to transport CO_2 in the volumes expected for Europe's 2050 decarbonization goals. Their findings justify the use of pipelines in this scenario, as they represent the most efficient long term transport option for continuous high volume CO_2 flows. For this case, every connection presented in figure 3 will become an offshore pipeline that exceeds the ship connection, which will become an offshore pipeline.

For the Hagenholz case study the calculation of the emissions of the different options was also computed following the analysis in section 2.4, with the $M_{\mathrm{CO2,fail}}$ being the one calculated using the Eq. 38.

3 Results

This section presents the main results of the economic assessment of resilience options within the CCTS chain, focusing on how their feasibility varies across a range of operational and system parameters. Rather than relying on single point cost estimates, the analysis evaluates the performance of each resilience strategy under multiple parameter combinations to capture the inherent uncertainty associated with long term CCTS deployment.

As described in Section 2.1, the resilience measures considered, differ in both their technical configuration and cost structure, leading to heterogeneous sensitivities to key parameters. These include the duration of disruptions affecting permanent transport connections (Failure Days), the CO_2 output of the WtE facility, transport distances, and the year of analysis. Variations in these parameters result in a wide range of cost outcomes, and their relative influence differs across resilience options.

The results are therefore presented in a comparative manner, systematically exploring how changes in individual parameters and their combinations affect the economic viability of each strategy. This approach enables the identification of cost drivers, thresholds, and parameter regimes under which specific resilience options become economically attractive or unattractive. The findings provide a basis for assessing the robustness of alternative resilience measures and support strategic decision making for resilient CCTS infrastructure planning.

3.1 Variation of the length of the connection

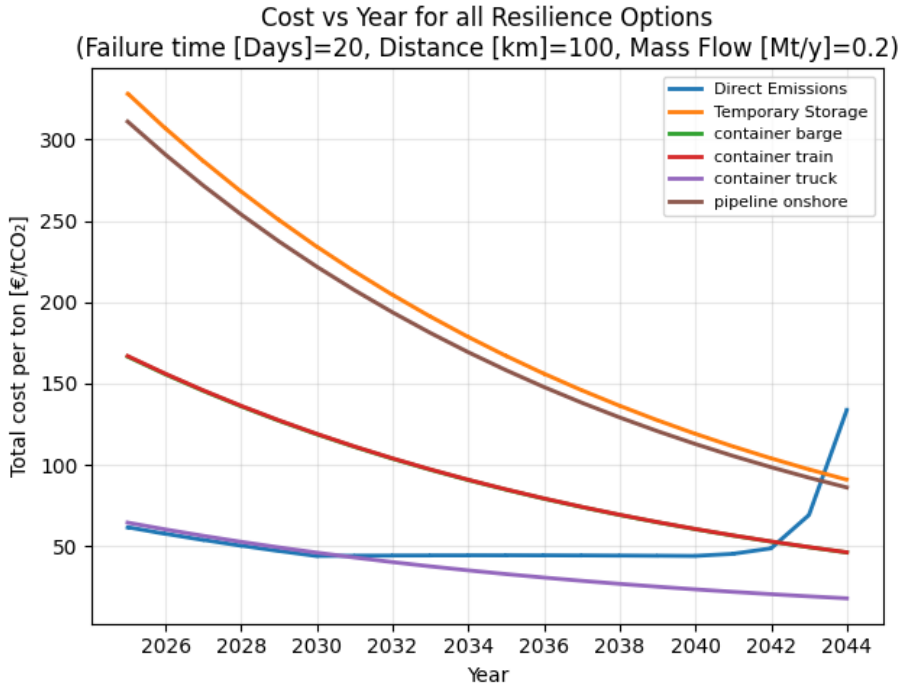
Transport distance is a key determinant of the economic performance of onshore resilience options, as it directly influences the scale of transport infrastructure and associated operational costs. This subsection examines how variations in connection length affect the relative cost competitiveness of onshore based resilience strategies, thereby identifying distance regimes in which specific options are economically favorable.

Figures 4a–4b present the evolution of costs for onshore resilience options at representative connection lengths of 100 km and 1000 km, respectively. Although the dominant visual feature is the temporal decline in costs, the results are interpreted primarily in terms of relative cost levels and their sensitivity to transport distance.

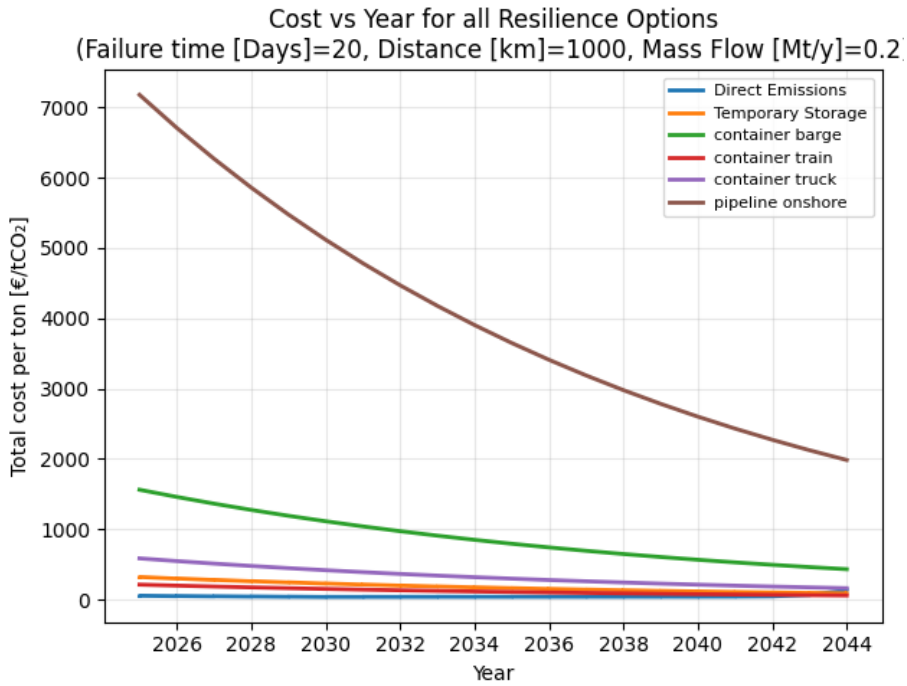
A pronounced distance dependency is observed for all onshore options. Increasing the connection length from 100 km to 1000 km results in a significant increase in the cost per ton of CO_2 for pipeline and container based transport solutions, reflecting the scaling of capital investment and operating costs with distance. In contrast, temporary storage remains largely unaffected by transport length, as its cost is driven mainly by storage capacity requirements rather than transport infrastructure. The only distance-related contribution arises from the marginal oversizing of the pipeline, which remains limited due to the relatively small additional CO_2 volumes associated with disruption events.

At shorter distances (Figure 4a), pipelines and container-based transport options are cost competitive with temporary storage throughout the entire time frame examined. In this configuration, the container barge option lies below the container train curve and is therefore not clearly distinguishable in the figure. It must be noted that for small distances the first technology to become economically feasible compared to directly emitting CO_2 in the atmosphere is container truck, which becomes cheaper very early in the time frame at 2030. At longer distances (Figure 4b), transport related costs increase sharply, particularly for pipeline based solutions, rendering onshore transport options significantly more expensive than temporary storage across the entire analysis period. Under these conditions, container train emerges as the economically dominant resilience option, with temporary storage having a very similar cost per ton, especially later in the time frame

Supplementary figures illustrating the cost evolution of all resilience options and parameter combinations are provided in Appendix D.



(a) Total cost per ton evolution throughout the years for 100 km length of connection



(b) Total cost per ton evolution throughout the years for 1000 km length of connection

Figure 4: Cost per ton evolution throughout the years for a constant length of connection

3.2 Dependency of resilience selection on the annual CO_2 mass of the WtE power plant

The annual mass flow of CO_2 produced by the WtE facility is a central parameter in determining the relative cost performance of resilience options, as it directly governs the total quantity of CO_2 that must be managed during disruption events. This subsection examines how changes in annual mass flow affect the cost competitiveness of alternative onshore resilience strategies relative to direct emissions, highlighting the conditions under which specific options become preferable.

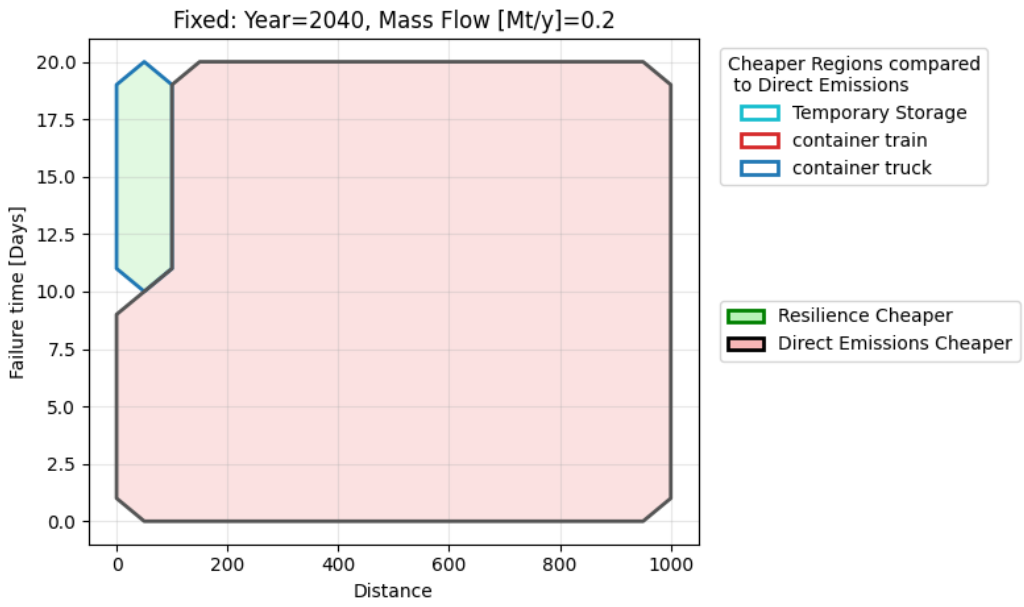
Within the modeling framework described in Section 2.1, transport and storage infrastructures are sized according to the assumed mass flow level. As a result, increasing the mass flow leads to a moderate increase in absolute system costs due to larger infrastructure requirements. However, the corresponding cost per ton of CO_2 decreases, as fixed capital and operational costs are distributed over a larger transported volume and the infrastructure operates closer to its design capacity. This effect is common across all resilience options and underpins the observed shift in relative cost performance with increasing mass flow.

Figures 5 illustrate how the regions in which resilience options are cheaper than direct emissions evolve with mass flow, focusing on the year 2040, which is characterized by a relatively high CO_2 tax and therefore provides a representative basis for comparison. The figures present only onshore transport and storage options.

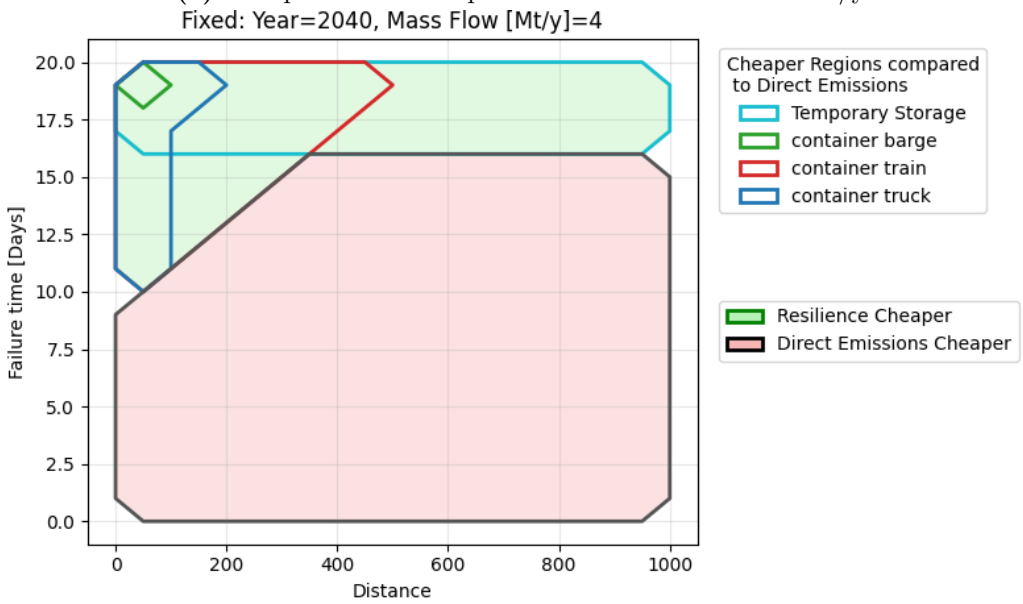
At low mass flow rates (Figure 5a), direct emissions remain the lowest cost option across most of the parameter space. Only container trucks become cheaper than direct emissions for short distances and longer failure durations, reflecting their relatively low capital intensity and flexibility at small volumes. This outcome is consistent with the techno economic formulation, which penalizes capital intensive resilience measures when utilization is limited.

At higher mass flow rates (Figure 5b), multiple resilience options become cheaper than direct emissions across substantial portions of the distance to failure duration space. Temporary storage dominates a large share of the parameter space, particularly for longer failure durations, where its ability to decouple capture from transport provides a cost advantage. Container trains and barges also become cheaper than direct emissions at increased mass flows, primarily at shorter transport distances. This reflects the steep scaling of transport costs with distance for both truck and train based solutions, while the transported mass during failure events remains constant.

Results for all resilience options for the higher tax scenario (year 2044) are provided in Appendix D.2. In all cases, the cost per ton of each resilience strategy is directly compared against the corresponding direct emissions case, ensuring consistent interpretation across parameter combinations.



(a) Cheapest resilience options for a mass flow of 0.2 Mt/y



(b) Cheapest resilience options for a mass flow of 4 Mt/y

Figure 5: Cheapest resilience option compared to the direct emissions case, for the same year and two different mass flows of the WtE power plant

3.3 Significance of the amount of days that the main connection has failed throughout the year

As discussed in Section 2.3, the number of failure days per year directly influences the utilization of resilience infrastructure and therefore has a strong impact on the capital cost per ton of CO_2. Failure duration is thus a key parameter in evaluating the economic performance of resilience strategies, particularly when considered in combination with transport distance and mass flow.

Figure 6 presents two parameter space comparisons at a fixed onshore transport distance of 500 km. Figure 6a represents a low frequency disruption scenario with 0.1 failure days per year, while Figure 6b illustrates a high impact scenario with 20 failure days in a year. Only onshore transport options are

considered for results clarity.

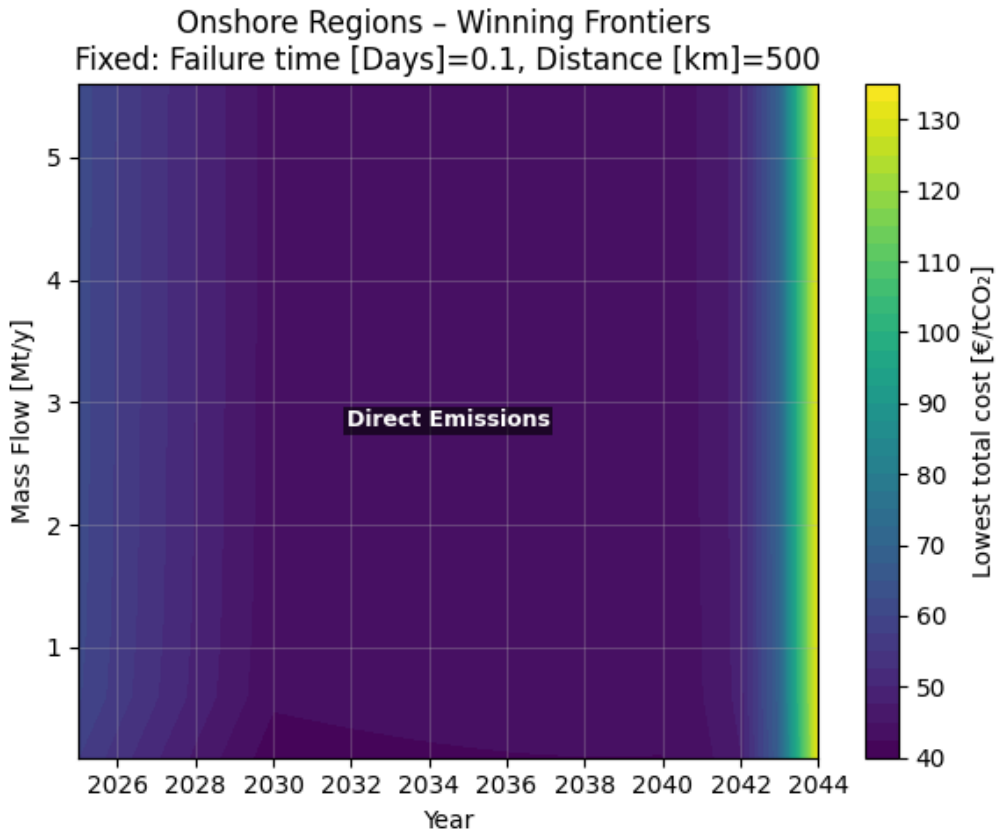
The choice of 0.1 failure days reflects the low expected probability of outages for permanent transport infrastructure, as discussed in Section 2.5. The 20 day scenario is included to represent rare but severe disruption events, such as the failure of a major transport route requiring an extended repair period before normal operation can resume.

At the fixed distance of 500 km, a clear distinction emerges between low and high failure duration regimes. Under low failure conditions, none of the resilience options achieve lower costs than direct emissions across the parameter space considered. The short duration of disruptions results in insufficient use of resilience infrastructure to offset associated capital investments, making direct emissions the cost minimizing option.

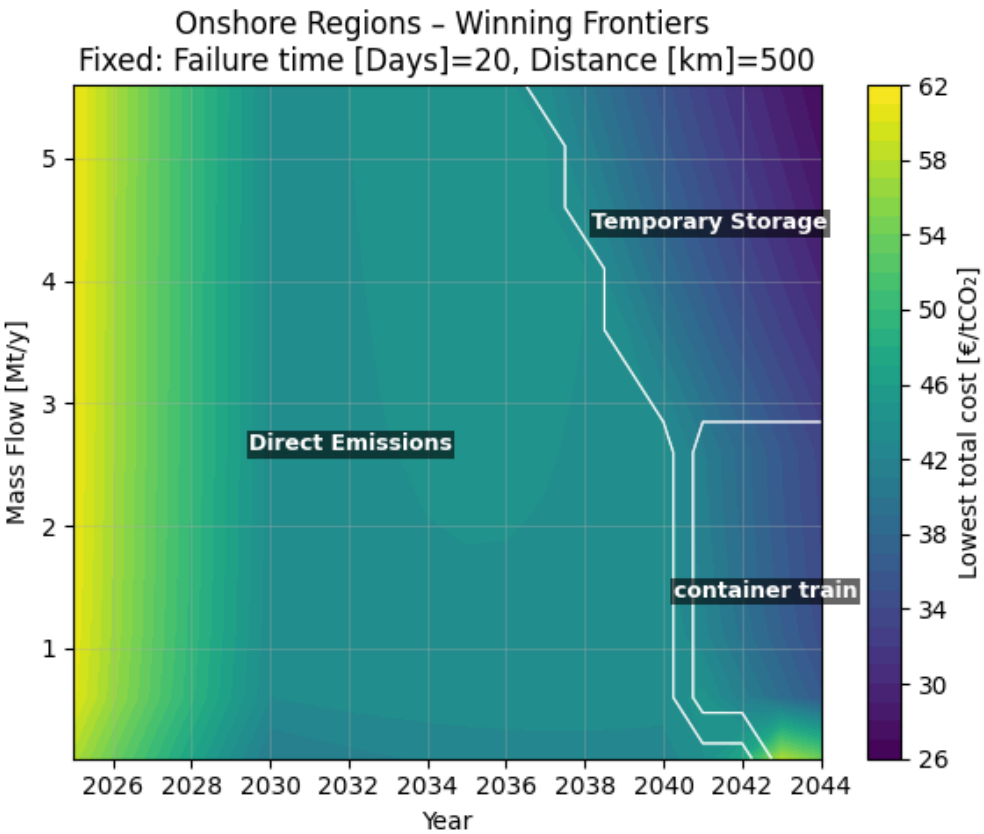
Under high failure durations, resilience strategies become cost competitive relative to direct emissions across specific combinations of mass flow and year. Temporary Storage achieves the lowest overall cost levels and dominates at higher mass flows, where its capital costs can be distributed over larger volumes of CO_2. At the same fixed distance, container train transport becomes cheaper than direct emissions at lower mass flow rates, where transport capacity requirements remain moderate and the cost advantage of Temporary Storage has not yet materialized. This behavior is consistent with the previous distance and mass flow analysis, in which container train based transport remains competitive at intermediate distances but loses its cost advantage at longer transport lengths. Compares to the mass flow analysis container trucks do not appear at all, due to the distance selected, which is much higher than the cost competitive region of the technology.

Overall, while Temporary Storage represents the most economical resilience option under high failure durations and high utilization, container train transport emerges as an additional onshore resilience strategy that becomes preferable to direct emissions under the combined conditions of prolonged disruptions, intermediate transport distance, and lower throughput levels.

Results for offshore configurations are provided in Appendix D.3.



(a) Low failure time throughout the year for moderate distance of 500 km



(b) High failure time throughout the year for moderate distance of 500 km

Figure 6: Combination of parameters that make each option the cheapest

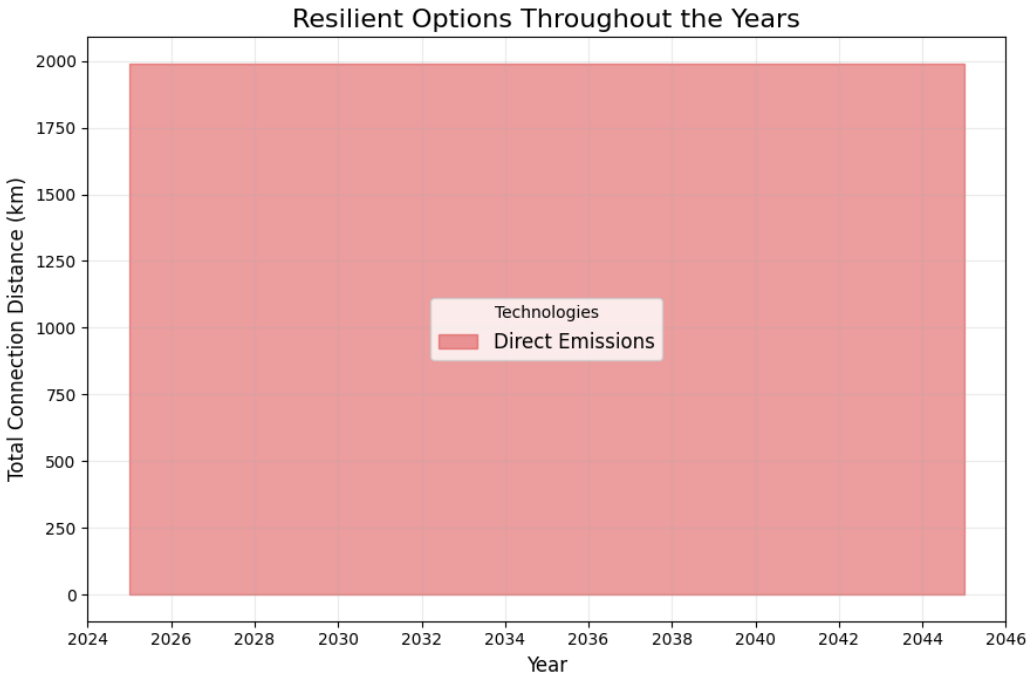
3.4 Hagenholz case study

To assess the practical implications of the parametric results, the resilience framework is applied to the Hagenholz WtE plant as a representative case study. Two alternative main connection configurations are examined, as described in Section 2.5: (i) the network layout proposed by (Burger et al. 2024), and (ii) a hypothetical configuration in which all long distance main connections are assumed to be pipelines of equal length. This comparison allows the influence of network topology and infrastructure type on resilience feasibility to be evaluated in a system specific context.

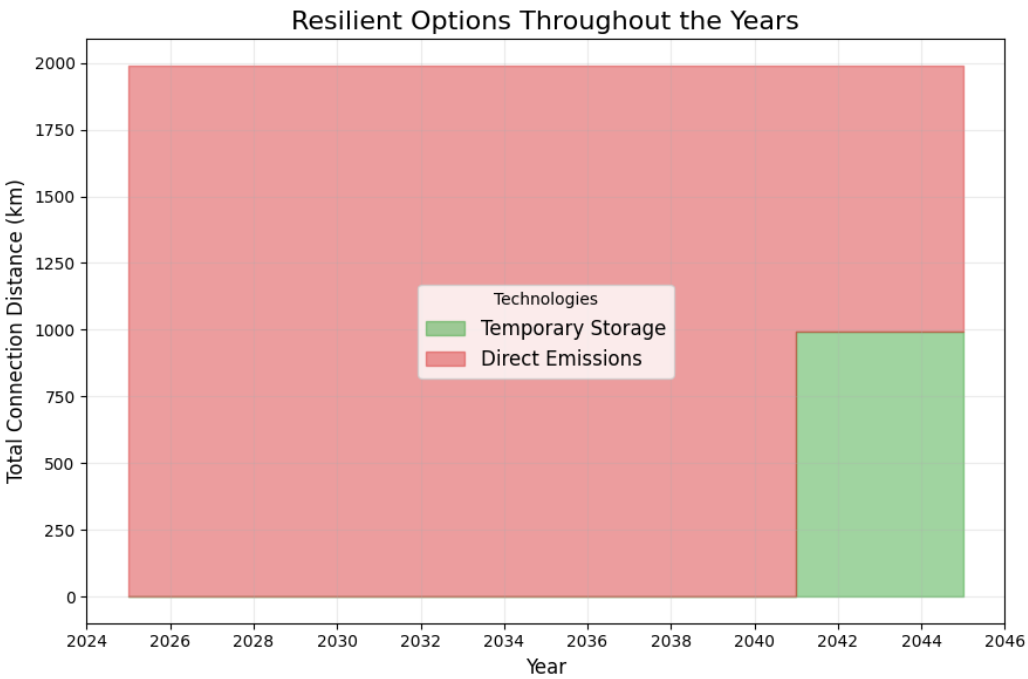
Figure 7 summaries the total length of the main connection that can be economically substituted by a resilience option over time. In the configuration proposed by (Burger et al. 2024) (Figure 7a), none of the main connections are economically attractive for a resilience strategy. This result is primarily driven by the relatively small mass of CO_2 produced by the power plant, combined with the low expected frequency of connection failures, which occur only once every several years. As discussed in the previous results sections, the combination of limited failure durations and low mass flow rates prevents alternative transport options from becoming economically viable in any year. Furthermore, even for short distance connections, the extremely small number of failure days per year means that container truck transport does not become cost effective, which was illustrated in Figure 5a, even for small mass flow of the WtE power plant.

A very different outcome emerges in the second configuration, where all main connections are modeled as pipelines (Figure 7b). In this scenario, temporary storage becomes the preferred resilience option, outperforming both container shipping and offshore pipeline redundancy across a substantial portion of the analysis period, in the 993 km offshore connection. Notably, Temporary Storage becomes cheaper than Direct Emissions in 2041. This result reflects its ability to decouple capture from transport during outages, avoiding the need for parallel long distance transport infrastructure. In a highly interconnected European pipeline network, this finding suggests that Temporary Storage could provide a particularly robust and cost effective resilience solution, both in terms of economic performance and deployment timing.

For shorter transport connections within the Hagenholz system, the results remain consistent with the broader parametric analysis presented earlier. Resilience options become economically attractive only under conditions of high failure duration or substantially increased mass flow. As neither of these conditions applies to the Hagenholz WtE plant under the assumptions considered, direct emissions remain the cost minimizing option for the shorter links.



(a) Connection suggested by (Burger et al. 2024)



(b) Connection consisting only of pipelines

Figure 7: Cheaper resilience option and distance that it can replace for the main connection suggested by (Burger et al. 2024) and a connection that is only consisting of pipelines

To complement the economic analysis, an emissions assessment is performed to evaluate the trade off between cost and climate impact associated with each resilience option. Figure 8 compares the total equivalent CO_2 emissions released to the atmosphere with the corresponding cost for the main connection identified in Figure 7. Figure 8 highlights the years in which each resilience option becomes cost competitive, enabling a direct comparison between emission reduction potential and economic feasibility.

Temporary storage emerges as both the earliest and the most impactful resilience option, achieving

substantial emission reductions at relatively low cost well before other technologies become competitive. The assumption made earlier for the emissions of the temporary storage to be equal to the ones of the pipeline, might not give the absolute correct emissions for the technology, but it does not materially affect the comparative outcome. Direct Emissions result in orders of magnitude higher total CO_2 emissions than any resilience alternative. Consequently, the dominance of Temporary Storage in the cost emissions space is driven primarily by the extreme emission burden associated with captured venting CO_2, rather than by specific modeling choices related to storage design.

Container shipping appears as a strong secondary option in cases where Temporary Storage is not available, offering meaningful emission reductions once it becomes economically viable.

Overall, the case study highlights a clear trade off between cost and emissions when selecting resilience strategies. While direct emissions may remain cost competitive under certain configurations, their associated climate impact is substantially higher than for all resilience options considered. The use of a logarithmic scale in Figure 8 further illustrates the magnitude of these differences and reinforces the importance of incorporating emissions considerations alongside economic criteria when evaluating resilience in CCTS systems.

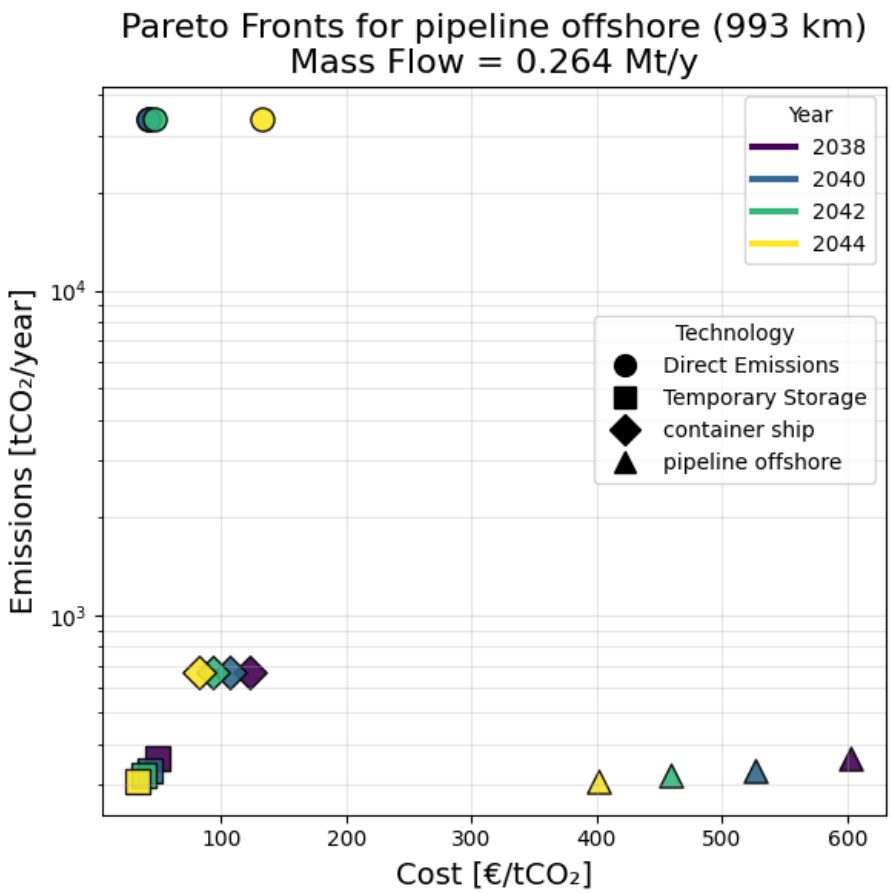


Figure 8: Emissions of different resilience options throughout the years compared to the cost of the option. For the main connection to be offshore pipeline of 993 km and the mass flow of the Hagenholz power plant

4 Conclusions and Outlook

4.1 Conclusion

This work developed a techno economic framework to evaluate resilience strategies in CO_2 transport within the CCTS chain, comparing redundant transport options and temporary CO_2 storage against the baseline of direct atmospheric release during main transport failure. The model integrates technology specific failure probability, dynamic carbon taxation, mass flow, transport distance, and outage duration, enabling a multidimensional economic assessment of resilience under realistic operational conditions.

The results demonstrate that resilience is not universally cost optimal, instead, feasibility emerges only within distinct regions of the parameter space, while also every resilient option has its own region of dominance. Among all studied inputs, the number of annual failure days and the CO_2 mass flow have the greatest influence on economic viability. High mass flow reduces the cost per ton by spreading infrastructure expenditure across more CO_2 capacity that need to be transported, while long outage durations increase the cost effective resilience options, since each alternative is utilized for more capacity and therefore the capital cost per ton is decreased. In contrast, transport distance mainly affects the choice of technology rather than the existence of feasibility itself, making short to medium transport distances more attractive container based transport and long distances favorable mostly to temporary storage, which reflects the independence of buffer storage to the length of the connection. Rising carbon taxation shifts feasibility earlier in time, illustrating that resilience is a time dependent decision.

Across most favorable scenarios, Temporary Storage emerges as the most competitive resilience solution, especially in the offshore region, particularly when outages are prolonged and plant output is high. Container truck becomes a good alternative earlier than other onshore options, however container train become viable for a larger set of parameters with the increase of the tax cost, especially for higher mass flow. Container ships appear as a strong offshore alternative when storage is unavailable and for low to moderate lengths, while offshore pipelines require extreme failure duration and high mass flow to outperform direct emissions, and therefore represent a niche rather than generalizable solution.

A critical insight from this study is the emissions cost trade off. Direct emissions remain economically favorable under present conditions, yet the associated climate burden is substantial. Delaying resilience implementation until cost break even potentially beyond 2040, may result in large cumulative CO_2 releases during failure events, since directly emitting CO_2 in the atmosphere is 1.5 to 2 orders of magnitude larger than any other resilience option. Thus, decision makers face a strategic balance, between short term cost minimization versus long term climate responsibility.

Overall, the conclusions show that resilience in CO_2 transport is a conditional investment. It provides substantial benefit only when system scale and operational risk exceed threshold values. Otherwise, direct emissions remain more economically attractive. However, under rising carbon prices and increasing deployment of large scale CCTS, resilience is likely to become both economically rational and environmentally essential.

4.2 Outlook

The presented framework provides an initial basis for resilience evaluation in CO_2 transport systems, but it also opens several opportunities for further development. First, extending the model beyond a single source, single route structure to a multi node network would allow the assessment of infrastructure sharing, alternative path routing, and cross border balancing. With increasing CCTS deployment, such network based perspectives will become increasingly relevant.

Failure modeling could be expanded beyond fixed Failure Day assumptions toward probabilistic or time-

varying outage representations. Dynamic switching between resilience options, or partial failure scenarios where capacity is weakened rather than lost, would more closely reflect real operational behavior. Integration with real time dispatch optimization could further support operational decision making in CCTS networks.

Future analyses should examine policy mechanisms intended to accelerate resilience adoption before economic break even. The results of this study suggest that small financial incentives may shift resilience adoption forward by several years, preventing large cumulative emissions during early deployment phases.

Taken together, these directions provide a pathway toward more comprehensive planning tools for future large scale CO_2 transport. As carbon prices rise, failure risk persists, and network scale increases, resilience is expected to transition from a marginal consideration to a core component of CCTS system design.

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1.15

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Appendices

A Trucks and trains model parameters

Below, all the data used for the creation of the train and truck model is presented, as well as the conversion factor that was used for transforming USD to EUROS. Table 3 includes all the parameters used in the modeling of the trucks used to transport the buffer capacity of CO_2, while table 4 has all the data used for the modeling of the trains.

Table 3: Key parameters and symbols for dedicated CO_2 truck transport.

Symbol	Parameter	Value	Source
P_{diesel}	Diesel price	1.55 €/L	(Cargopedia SRL 2025)
c_f	Average fuel consumption	35 L/100 km	(Dewhurst 2024)
v	Average speed	60 km/h	(Oeuvray et al. 2024)
c_m	Maintenance cost coefficient	0.10 €/km	(Myers et al. 2024)
c_t	Tire cost coefficient	0.02 €/km	(Myers et al. 2024)
c_i	Insurance cost coefficient	0.05 €/km	(Myers et al. 2024)
$C_{\text{driver,h}}$	Driver hourly wage	18.2 €/h	(Myers et al. 2024)
Cap_{truck}	Truck CO_2 capacity	26 t	(Oeuvray et al. 2024)
t_{load}	Loading/unloading time	1 h	(Oeuvray et al. 2024)
t_{border}	Border delay	0.5 h	(Oeuvray et al. 2024)
t_{break}	Driver rest time	0.2 h	(Oeuvray et al. 2024)
$C_{\text{truck,veh}}$	Truck cost	202,221 USD	(Myers et al. 2024)
C_{trailer}	Trailer cost	331,671 USD	(Myers et al. 2024)
n	Truck lifetime	15 years	(Myers et al. 2024)

Table 4: Key parameters and symbols for dedicated CO_2 train transport.

Symbol	Parameter	Value	Source
v	Average speed	70 km/h	(Oeuvray et al. 2024)
t_{load}	Loading time	18 h	(Oeuvray et al. 2024)
t_{unload}	Unloading time	18 h	(Oeuvray et al. 2024)
N_{wagon}	Wagons per train	20	(Oeuvray et al. 2024)
Cap_{wagon}	Wagon capacity	50 t	(Oeuvray et al. 2024)
C_{loco}	Locomotive cost	4,000,000 USD	(Myers et al. 2024)
C_{wagon}	Wagon cost	239,347 USD	(Myers et al. 2024)
C_{OPEX}	Base OPEX per train·km	25 €/train·km	(Myers et al. 2024)
m_{maint}	Annual maintenance fraction	0.03	(Myers et al. 2024)
n_L	Locomotive lifetime	25 years	(Myers et al. 2024)
n_W	Wagon lifetime	30 years	(Myers et al. 2024)

Additionaly, supporting equations for cost calculation of different parameters for trucks are presented in equations 39-42, while all costs are expressed in euros using a conversion factor, as of the time that the report was written with USD being equal to 0.86 Euros.

$$C_{\text{fuel}} = P_{\text{diesel}}\, c_f\, \frac{2L}{100} [€] \tag{39}$$

$$C_{\text{maint}} = c_m(2L)[€] \tag{40}$$

$$C_{\text{tires}} = c_t(2L)[€] \tag{41}$$

$$C_{\text{ins}} = c_i(2L)[€] \tag{42}$$

B Parameters used for the modelling of barges, ships and pipelines

This Appendix includes all the parameters used for equations 20 and 21, which represent the model econometric model for barges and ships and the model for pipelines presented in (Oeuvray et al. 2024).

Table 5: Model parameters for dedicated barge transport (Oeuvray et al. 2024).

Symbol	Parameter	Value	Unit
a_1	Fixed cost coefficient	0.03	€/t·km
a_2	Distance term coefficient	4.66	€/t
a_3	Flow term coefficient	5190	€/km·y
f_{OPEX}	OPEX fraction	0.15 (Vikara et al. 2025)	–

Table 6: Model parameters for dedicated ship transport (Oeuvray et al. 2024).

Symbol	Parameter	Value	Unit
a_1	Fixed cost coefficient	0.01	€/t·km
a_2	Distance term coefficient	9.95	€/t
a_3	Flow term coefficient	3280	€/km·y
f_{OPEX}	OPEX fraction	0.15 (Vikara et al. 2025)	–

Table 7: Model parameters for onshore and offshore pipeline transport (Oeuvray et al. 2024)

Symbol	Parameter	Onshore	Offshore	Unit
a_1	Fixed cost coefficient	0.02	0.02	€/t·km
a_2	Scaling factor	260	58.3	€/t·km
a_3	Distance exponent	0.07	0.07	–
a_4	Flow exponent	-0.61	-0.51	–
f_{OPEX}	OPEX fraction	0.026 (Solomon et al. 2024)	0.026 (Solomon et al. 2024)	–
m_{ref}	Reference flow	1	1	$MtCO_2$/y
d_{ref}	Reference distance	1	1	km

C Hagenholz case study constraints and route

Each main connection type is constrained depending the alternatives that are feasible for each case, the set of resilience options available for each main connection is:

$$\text{pipeline onshore: } \{\text{truck, train, pipeline onshore, Temporary Storage, Direct Emissions}\},$$
$$\text{container truck: } \{\text{truck, train, pipeline onshore, Direct Emissions}\},$$
$$\text{container train: } \{\text{truck, train, pipeline onshore, Direct Emissions}\},$$
$$\text{container barge: } \{\text{barge, truck, train, pipeline onshore, Direct Emissions}\},$$
$$\text{container ship: } \{\text{ship, pipeline offshore, Direct Emissions}\},$$
$$\text{pipeline offshore: } \{\text{ship, pipeline offshore, Temporary Storage, Direct Emissions}\}.$$

The type of main connections used for the case study, as well as the lengths and locations are presented in table 8.

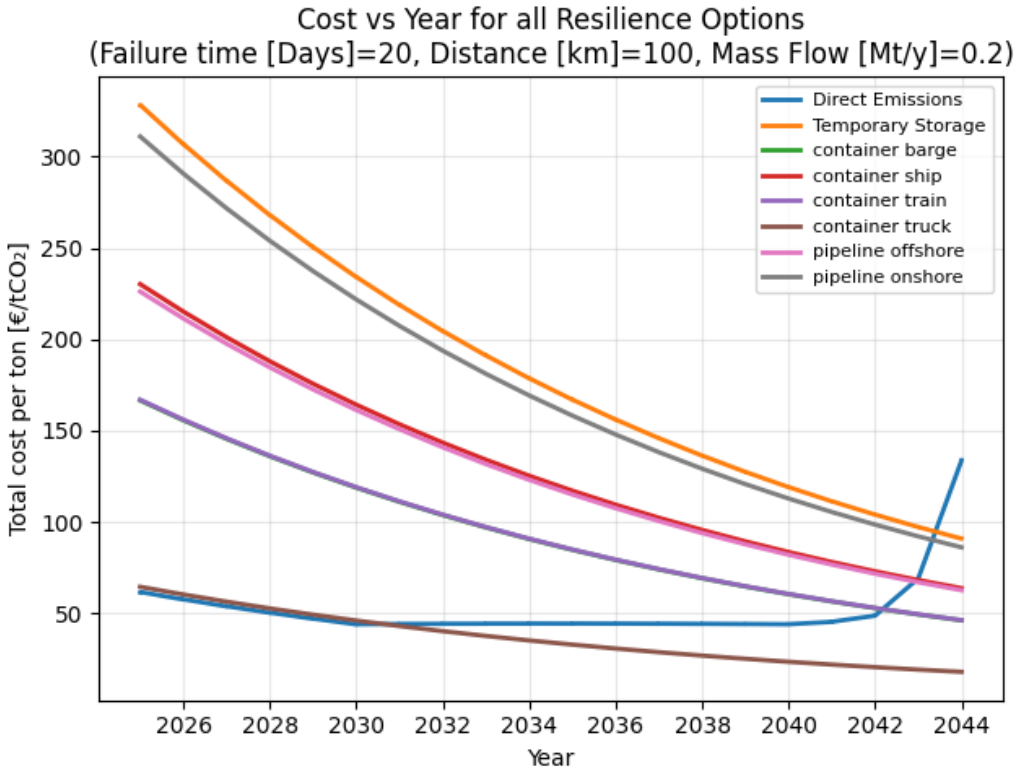
Route	**Distance (km)**
KVA Hagenholz → ARA Werdhölzli (pipeline onshore)	13
ARA Werdhölzli → Basel (container truck)	80
Basel → Rotterdam (container barge)	853
Rotterdam → Bergen (container ship)	993
Bergen → Northern Lights (Øygarden) (container truck)	51

Table 8: Transport routes for the Hagenholz case study

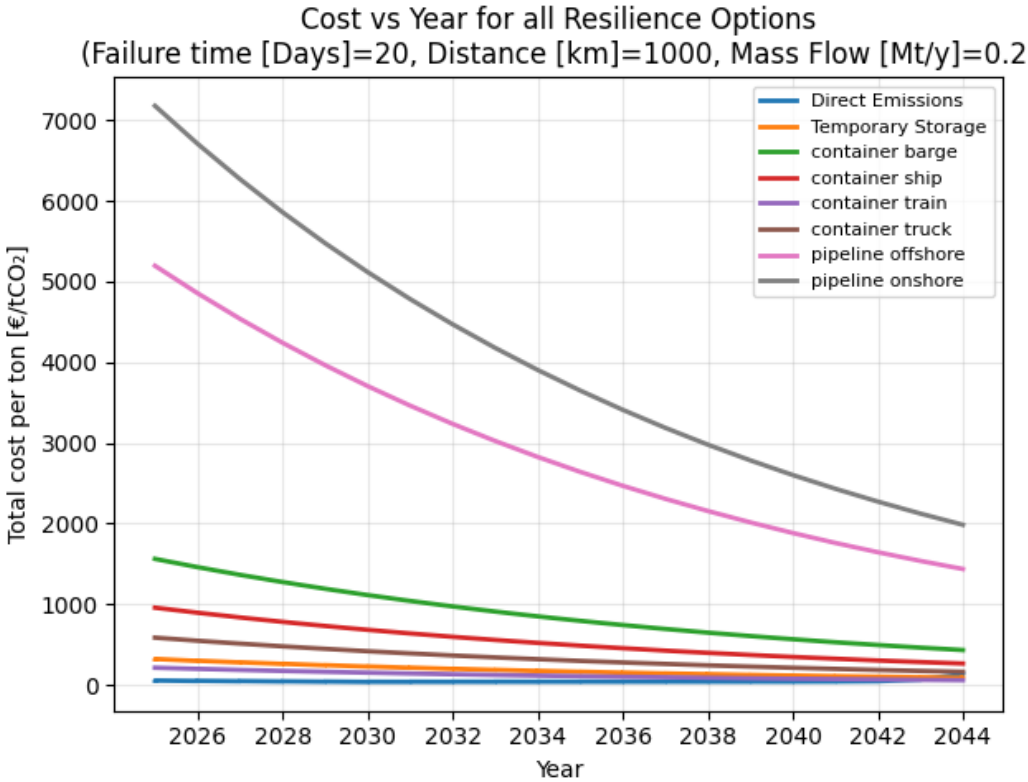
D Supplementary graphs for the results section

D.1 Variation of connection lengths

Figures 9a and 9b present the results of the cost variation over the years for all resilience options at distances of 100 and 1000 km, respectively. Both graphs show that a wider range of options are economically comparable with Direct Emissions at lower distances, since increasing the length of the connection increases the capital investment of most of the options.



(a) Length of connection is fixed at 100 km



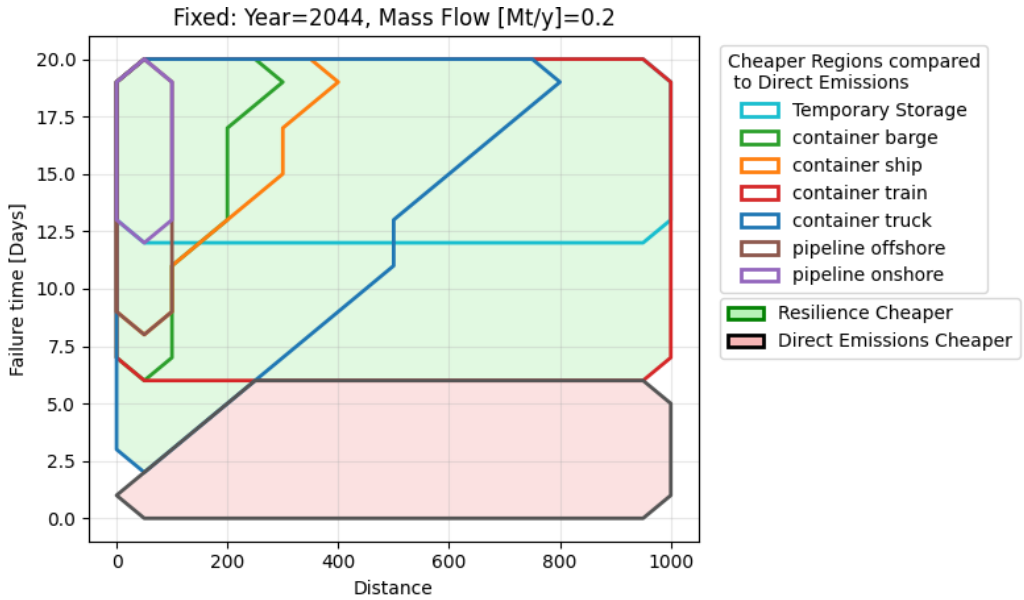
(b) Length of connection is fixed at 1000 km

Figure 9: Comparison of the cost evolution throughout the years for two different connection lengths for all resilience options

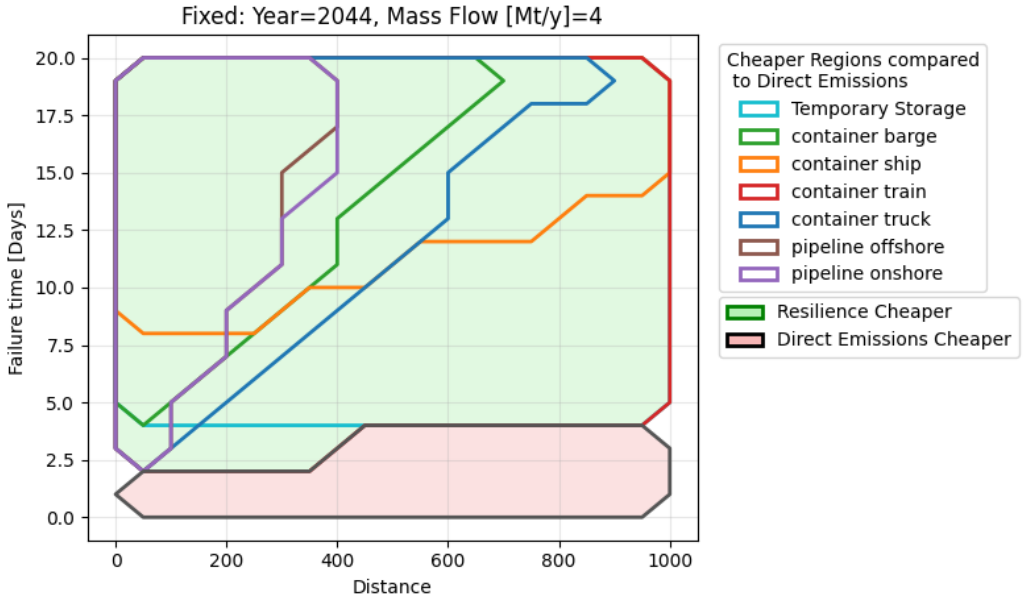
D.2 Dependency on the CO_2 mass flow of the WtE power plant

In Figure 10, the mass flow comparison is repeated for the year 2044, a period characterized by a substantially higher CO2 tax than in 2040. Under these conditions, a larger range of resilience solutions becomes cost competitive relative to the Direct Emissions case. Both offshore pipelines and shipping alternatives emerge as economically feasible for certain combinations of parameters, in low and high mass flow scenarios.

For small mass flows, several options appear competitive across a wider range of transport distances. However, economic feasibility continues to be based on a relatively long duration of failure. This requirement remains a limiting factor, and for much of the parameter space, Direct Emissions continue to dominate. However, the shift compared to 2040 is evident, as carbon pricing increases, the thresholds at which resilience becomes favorable decrease, allowing more technologies to enter cost competitive regions.



(a) Cheapest resilience options for a mass flow of 0.2 Mt/y in 2044



(b) Cheapest resilience options for a mass flow of 4 Mt/y in 2044

Figure 10: Cheapest resilience option compared to the direct emissions case

D.3 Significance of the amount of days that the main connection has failed throughout the year

Figure 11 demonstrates that, under conditions of prolonged failure duration, offshore pipelines become the most cost effective option compared to Direct Emissions across a large portion of the parameter space. In this scenario, container ships do not appear favorable, which aligns with earlier results, indicating that shipping only becomes competitive at shorter transport distances. The results for 0.1 failure days are identical to those presented in figure 6a

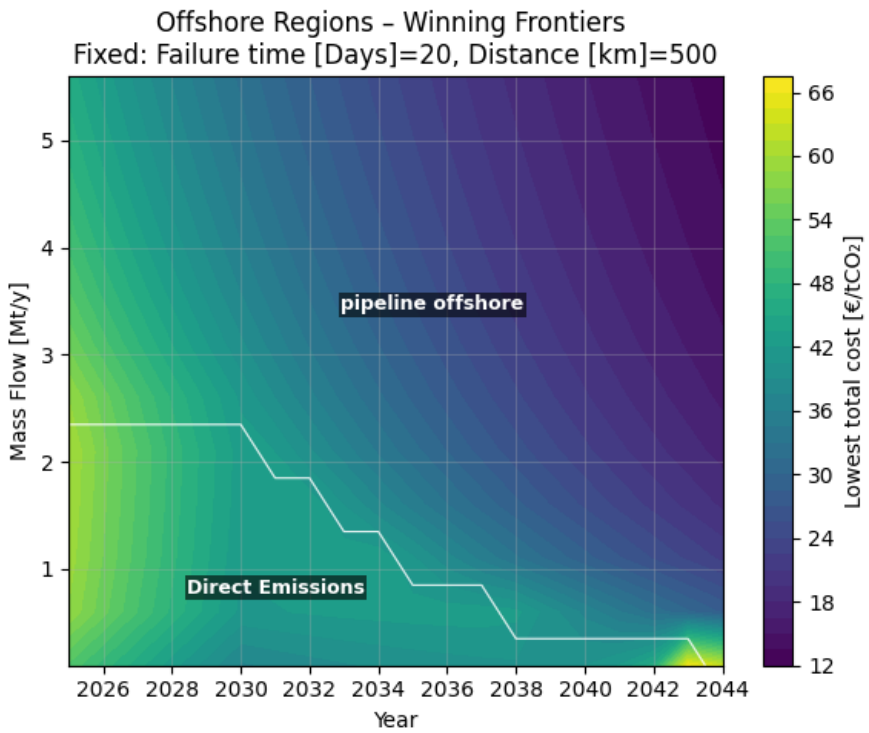


Figure 11: Set of parameters where each offshore resilience option becomes cost viable